

# DISCUSSION NOTE: TWO-PHASE DEAD CORE FOR THE FRACTIONAL $p$ -LAPLACIAN STATUS, ONE QUESTION ABOUT *IMPROVED REGULARITY FOR A NONLOCAL DEAD-CORE PROBLEM*, AND A PROPOSED ARGUMENT

HIKMATULLO ISMATOV

## 1. STATUS IN ONE PARAGRAPH

For  $(-\Delta_p)^s u = u_+^\gamma - u_-^\gamma$  in  $B_1$  ( $0 < \gamma < p-1$ ,  $p \geq 2$ ), the target is growth of order  $\alpha = \frac{sp}{p-1-\gamma}$  at points with  $u = |Du| = 0$ . The scaling exponent and the vanishing of the right-hand side along blow-ups (via  $(p-1)$ -homogeneity, replacing linearization) are done. Following Correa–dos Prazeres §6, the Liouville theorem is reformulated with a vanishing 1-jet hypothesis, avoiding affine subtraction; the jet-to-growth lemma transplants verbatim. The interior  $C^{1,\alpha_0}$  input is Giovagnoli–Jesus–Silvestre (arXiv:2509.26565); their own remark suggests the needed  $f \in L^\infty$  extension is feasible for  $p < \frac{1}{1-s}$  (forcing  $s > \frac{1}{2}$ , as in your paper). Two caveats found: their constants blow up as  $s \nearrow 1$ , so the local two-phase theorem should be proved directly rather than by the  $s \nearrow 1$  limit; and the tail bookkeeping below. With these ingredients the **main growth estimate is proved modulo one Liouville statement** (contradiction + a  $\theta_k$  blow-up run at a generic sub-sharp exponent, the absorption killed by homogeneity, the limit killed by the Liouville). The only open inputs are that Liouville in the nonlocal setting and a nondegeneracy property; see §4.

## 2. THE QUESTION ABOUT (4.12)

In the proof of Theorem 4.1, the growth bound (4.11) for  $w_k = v_k(\cdot + h) - v_k(\cdot)$  rests on (4.8), which is justified for  $|x| \leq \frac{1}{2r_k}$  (as (4.7) requires  $r_k R < 1/2$ ). Beyond that radius,  $u_k$  is only bounded, so  $|w_k| \leq 2(\theta_k(r_k)r_k^{\frac{2s}{1-\gamma}})^{-1}$ . Passing (4.12) to the limit via [CS, ARMA 2011] requires  $w_k \rightarrow w_0$  in  $L^1(\mathbb{R}^n, \omega)$  *globally* (their Remark 4.4 shows local convergence cannot suffice), and the outer region contributes  $\sim \theta_k^{-1} r_k^{-\frac{2s\gamma}{1-\gamma}}$ , which is bounded only under a lower bound on  $\theta_k$  that (4.5) does not seem to provide. **Question: how is the far field handled in (4.12)?** I may well be missing something standard.

## 3. A PROPOSED ARGUMENT (CHECKED STEP BY STEP; SOUND IN OUTLINE)

Truncate:  $\tilde{w}_k = w_k \chi_{B_{\rho_k}}$ ,  $\rho_k = \frac{1}{4r_k}$ . Then  $(-\Delta)^s \tilde{w}_k = f_k + H_k$  in  $B_R$ , where  $H_k$  is the far field as a datum. The point is that only the *oscillation* of  $H_k$  matters:

- (1)  $\|\nabla H_k\|_{L^\infty(B_R)} \lesssim C_h \rho_k^{\alpha-2s-2} + \theta_k^{-1} r_k^{2s+1-\alpha} \rightarrow 0$ , the second exponent being positive iff  $\gamma < \frac{1}{1+2s}$  (the standing assumption). Hence  $H_k = a_k + o(1)$ ,  $a_k$  constant.
- (2)  $a_k$  is bounded *a posteriori*:  $\tilde{w}_k$  has uniform local regularity and uniform  $L_s^1$  norm, so  $(-\Delta)^s \tilde{w}_k$  is uniformly bounded at a fixed point, while  $f_k \rightarrow 0$ .
- (3) Stability now applies and gives  $(-\Delta)^s(v_0(\cdot + h) - v_0) = a(h)$  with  $a(h)$  finite.

- (4) The cocycle identity  $w^{h_1+h_2} = w^{h_1}(\cdot + h_2) + w^{h_2}$  forces  $a(h) = b \cdot h$ .
- (5) Second differences:  $(-\Delta)^s \Delta_h^2 v_0 = 0$  entire, with growth  $(1 + |x|)^{\alpha-2}$ ; Theorem 3.1 ( $\nu = 1$ ) makes  $\Delta_h^2 v_0$  affine for every  $h$ , so  $v_0$  is a polynomial of degree  $\leq 2$  (by the growth  $\alpha < 1 + 2s$ ), and the vanishing 2-jet at the branching point gives  $v_0 \equiv 0$ .

No lower bound on  $\theta_k$  is needed. The step I worried about most is (2): it is *not* circular — the bound on  $(-\Delta)^s \tilde{w}_k$  at a fixed point splits into a local singular part (controlled by the uniform interior  $C^{2s+\gamma}$  estimate) and a tail (controlled using only the growth bound  $|w_k(y)| \leq C_h |y|^{\alpha-1}$  on the truncated support, finite iff  $\gamma < \frac{1}{1+2s}$ ), neither of which uses  $a_k$  or any bound on  $\theta_k$ ;  $a_k$  is then solved for. The truncation is precisely what manufactures the global  $L_s^1$  convergence that (4.12) needs.

The same far-field-as-datum mechanism carries to  $p \neq 2$  for steps (1)–(3): the increments solve the *linear* equation  $L_{K_{v_k}} w = g$  with the kernel  $K_v \simeq |v(x) - v(y)|^{p-2} |x - y|^{-n-sp}$  of GJS §7, and the oscillation computation produces the additional condition  $\gamma < \frac{p-1}{1+sp}$  (which coincides with  $\frac{1}{1+2s}$  at  $p = 2$ , stricter for  $p > 2$ ). Steps (4)–(5), however, use translation invariance of  $(-\Delta)^s$ , which the  $v$ -dependent kernel  $K_v$  does *not* have; so for  $p \neq 2$  I would drop the constant- $a(h)$  route and apply the vanishing-jet Liouville to  $v_0$  directly (where  $a(h)$  never appears) — this is one of the points I would like to confirm with you.

#### 4. WHAT I WOULD LIKE TO DISCUSS

- (1) Whether the (4.12) argument of §2–§3 is right, or whether the far field has a standard justification I missed.
- (2) Whether, for  $p \neq 2$ , the vanishing-jet Liouville on  $v_0$  indeed removes the need for steps (4)–(5) (which rely on the translation invariance lost when  $p \neq 2$ ).
- (3) **The nonlocal Liouville — which route?** Your §6 (and Correa–dos Prazeres) prove it by rescaling  $v$  and applying the sharp  $C^{1,\alpha_0}$  estimate to  $v_l = v(\cdot)/l^{1+\beta}$ . Nonlocally this seems blocked:  $v_l$  keeps the growth  $1 + \beta = \alpha$ , and  $\text{Tail}(v_l) < \infty$  would need  $\alpha(p-1) < sp$ , whereas  $\alpha(p-1) - sp = \frac{\gamma ps}{p-1-\gamma} > 0$  for every  $\gamma > 0$  (truncating leaves a datum with the same divergent exponent). So I expect one must pass to *increments*  $w^h$  (which lower the growth to  $\beta < 1$  and have finite tail under the tail condition), giving the cap  $\alpha < 2$  rather than  $\alpha < 1 + \alpha_0$  — with  $\alpha_0$  then needed only for compactness, not quantitatively. Two questions: (a) is the direct route really unavailable nonlocally, or is there a way to apply the estimate past the divergent tail? (b) for the increment route, the linearized kernel  $K_v \simeq |v(x) - v(y)|^{p-2} |x - y|^{-n-sp}$  degenerates where  $v$  is flat; is there a clean way to get *nondegeneracy* of the blow-up limit  $v_0$  (gradient not identically zero on open sets) so the kernel is uniformly elliptic? These two are now the only open inputs to the main theorem.
- (4) Proving the local two-phase  $p$ -dead-core theorem directly (the  $s \nearrow 1$  route is blocked by the non-uniformity of the GJS constants); it does not appear in the literature and may stand alone.
- (5) Whether to include the  $f \in L^\infty$  extension of GJS for  $p < \frac{1}{1-s}$  as a section of our paper.

All details, computations and sources are in the project repository: [github.com/hikmatulloismatov1999/fracti](https://github.com/hikmatulloismatov1999/fracti)